

คาบตัว CALCULUS 1 สัปดาห์ที่ 10

$$F(x) = x^2 + C$$

เพิ่มปฏิยานุพันธ์ของ
หว่าไป

$$f(x) = 2x$$

นิยามของปฏิยานุพันธ์ ให้ f เป็นฟังก์ชัน

ฟังก์ชัน $F(x)$ เป็น ปฏิยานุพันธ์ ของ $f(x)$ เมื่อ $F'(x) = f(x)$

นิยามของปริพันธ์ไม่จำกัดเขต

ให้ f เป็นฟังก์ชันและ $F'(x) = f(x)$ จะได้ว่า $\int f(x)dx = F(x) + C$

1. จงตรวจสอบว่า $F(x) = \frac{x^3 \ln x}{3} - \frac{x^3}{9}$ เป็นปฏิยานุพันธ์ของ $f(x) = x^2 \ln x$

$$F'(x) = f(x)$$

พิจารณา $F'(x) = \frac{1}{3} [(x^3)(\ln x)' + \ln x (x^3)'] - \frac{1}{9} \frac{dx^3}{dx}$

$$= \frac{1}{3} \left[\cancel{x^3}^2 \left(\frac{1}{\cancel{x}} \right) + \ln x (3x^2) \right] - \frac{1}{9} (\cancel{3x^3})^2$$

$$= \frac{\cancel{x^3}^2}{3} + \frac{3x^2 \ln x}{3} - \frac{\cancel{x^3}^2}{3}$$

$$= \frac{\cancel{3x^2}^2 \ln x}{\cancel{3}} = x^2 \ln x$$

เพื่อ $F'(x) = f(x)$

ดังนั้น $F(x)$ เป็นปฏิยานุพันธ์ของ $f(x)$

สูตรของปริพันธ์ไม่จำกัดเขต

$$1. \int c f(x) dx = c \int f(x) dx \Rightarrow \int 5x^2 dx = 5 \int x^2 dx$$

$$2. \int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

$$3. \int k dx = kx + C \Rightarrow \int 7 dx = 7x + C \quad \int -2 dx = -2x + C$$

$$4. \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad ; \text{เมื่อ } n \neq -1$$

$$5. \int \frac{1}{x} dx = \ln |x| + C$$

$$\int x^2 dx = \frac{x^{2+1}}{2+1} + C = \frac{x^3}{3} + C$$

$$\int dx = \int 1 dx = x + C$$

$$6. \int e^x dx = e^x + C$$

$$7. \int b^x dx = \frac{b^x}{\ln b} + C$$

$$8. \int \sin x dx = -\cos x + C$$

$$9. \int \cos x dx = \sin x + C$$

$$10. \int \sec^2 x = \tan x + C$$

$$11. \int \sec x \tan x dx = \sec x + C$$

$$12. \int \csc x \cot x dx = -\csc x + C$$

$$13. \int \csc^2 x dx = -\cot x + C$$

$$\int \frac{1}{x^2} dx = \int x^{-2} dx$$

$$14. \int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C = -\arccos x + C$$

$$15. \int \frac{1}{1+x^2} dx = \arctan x + C = -\operatorname{arccot} x + C$$

$$16. \int \frac{1}{|x|\sqrt{x^2-1}} dx = \operatorname{arcsec} x + C = -\operatorname{arccsc} x + C$$

$$17. \int \sinh x dx = \cosh x + C$$

$$18. \int \cosh x dx = \sinh x + C$$

$$2. \int (x^{1.3} + 7x^{2.5}) dx \quad \int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$= \int x^{1.3} dx + \int 7x^{2.5} dx$$

$$= \frac{x^{1.3+1}}{1.3+1} + C_1 + \frac{7x^{2.5+1}}{2.5+1} + C_2 = \frac{x^{2.3}}{2.3} + \frac{7x^{3.5}}{3.5} + C$$

$$3. \int (\sqrt{4}x^5) dx$$

$$\frac{x^6}{6} + C$$

$$4. \int (x^3 + 3x^2 - 5x + 3) dx$$

$$\frac{x^4}{4} + x^3 - \frac{5x^2}{2} + 3x + C$$

$$5. \int \left(5 + \frac{2}{3}x^2 + \frac{3}{4}x^3\right) dx$$

$$5x + \frac{2}{9}x^3 + \frac{3}{16}x^4 + C$$

$$6. \int (x+6)(x+2) dx = \int x^2 + 8x + 12 dx$$

$$= \frac{x^3}{3} + \frac{8x^2}{2} + 12x + C$$

$$= \frac{x^3}{3} + 4x^2 + 12x + C$$

$$7. \int \sqrt{x}(x^2 + 2x + 3) dx = \int x^{\frac{1}{2}}(x^2 + 2x + 3) dx$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$= \int x^{\frac{5}{2}} + 2x^{\frac{3}{2}} + 3x^{\frac{1}{2}} dx$$

$$= \frac{x^{\frac{7}{2}}}{\frac{7}{2}} + \frac{2x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{3x^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= \frac{2x^{\frac{7}{2}}}{7} + \frac{4x^{\frac{5}{2}}}{5} + \frac{6x^{\frac{3}{2}}}{3} + C$$

$$8. \int \left(x^2 + x + 1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^2 + 1}\right) dx$$

$$= \frac{x^3}{3} + \frac{x^2}{2} + x + \ln|x| + \frac{x^{-1}}{-1} + \arctan x + C$$

$$= \frac{x^3}{3} + \frac{x^2}{2} + x + \ln|x| - \frac{1}{x} + \arctan x + C$$

$$9. \int \frac{t\sqrt{t} + \sqrt{t}}{t^2} dt$$

$$2t^{\frac{1}{2}} - 2t^{-\frac{1}{2}} + C$$

$$10. \int \left(\frac{1+r}{r} \right)^2 dr = \int \frac{1+2r+r^2}{r^2} dr = \int \frac{1}{r^2} + \frac{2}{r} + 1 dr$$

$$-\frac{1}{r} + 2\ln|r| + r + C$$

$$11. \int (\sin x + \frac{1}{2}x + e^x) dx = -\cos x + \frac{1}{2} \frac{x^2}{2} + e^x + C$$

$$= -\cos x + \frac{x^2}{4} + e^x + C$$

$$12. \int \left(\frac{2}{5} \sec \theta \tan \theta \right) d\theta = \frac{2}{5} \int \sec \theta \tan \theta d\theta$$

$$= \frac{2}{5} \sec \theta + C$$

$$13. \int \left(-\frac{\sec^2 x}{3} \right) dx$$

$$= -\frac{1}{3} \tan x + C$$

$$14. \int \frac{2x^3 - x}{x^5} dx$$

$$\frac{1}{3x^3} - \frac{2}{x} + C$$

$$15. \int \frac{(\sqrt{3} - \sqrt{x})^2}{\sqrt{x}} dx$$

$$= \int \frac{3 + 2\sqrt{3}\sqrt{x} + x}{\sqrt{x}} dx$$

$$= \int 3x^{-\frac{1}{2}} + 2\sqrt{3} + x^{\frac{1}{2}} dx$$

$$= 6x^{\frac{1}{2}} + 2\sqrt{3}x + \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$16. \int \frac{dx}{x^2}$$

$$= -\frac{1}{x} + C$$

$$17. \int t(2t^2 + 1)^2 dt$$

$$\frac{2}{3} t^6 + t^4 + \frac{t^2}{2} + C$$

$$18. \int \frac{dx}{\sqrt[3]{x}}$$

$$\frac{3}{2} x^{\frac{2}{3}} + C$$

$$19. \int \left(x^3 - \pi + \frac{1}{8\sqrt{x^7}} \right) dx = \int x^3 - \pi + \frac{x^{-\frac{7}{2}}}{8} dx$$

$$\int k dx = Kx + C$$

$$-\frac{7}{2} + 1 = -\frac{5}{2}$$

$$= \frac{x^4}{4} - \pi x + \frac{1}{8} \frac{x^{-\frac{7}{2}+1}}{-\frac{7}{2}+1} + C$$

$$= \frac{x^4}{4} - \pi x - \frac{2x^{-\frac{5}{2}}}{40} + C$$

$$20. \int (2x + 1)(3 - x^2) dx$$

$$-\frac{x^4}{2} - \frac{x^3}{3} + 3x^2 + 3x + C$$

$$21. \int t^{-2}(2t^4 + 3t^2 - 2t^{-3}) dt$$

$$\frac{2}{3} t^3 + 3t + \frac{1}{2} t^{-4} + C$$

$$22. \int \sec \theta (\tan \theta - 2 \cos \theta) d\theta = \int \sec \theta \tan \theta - 2 \cos \theta \sec \theta d\theta$$

$$= \int \sec \theta \tan \theta - 2 \cancel{\cos \theta} \frac{1}{\cancel{\cos \theta}} d\theta$$

$$= \int \sec \theta \tan \theta - 2 d\theta$$

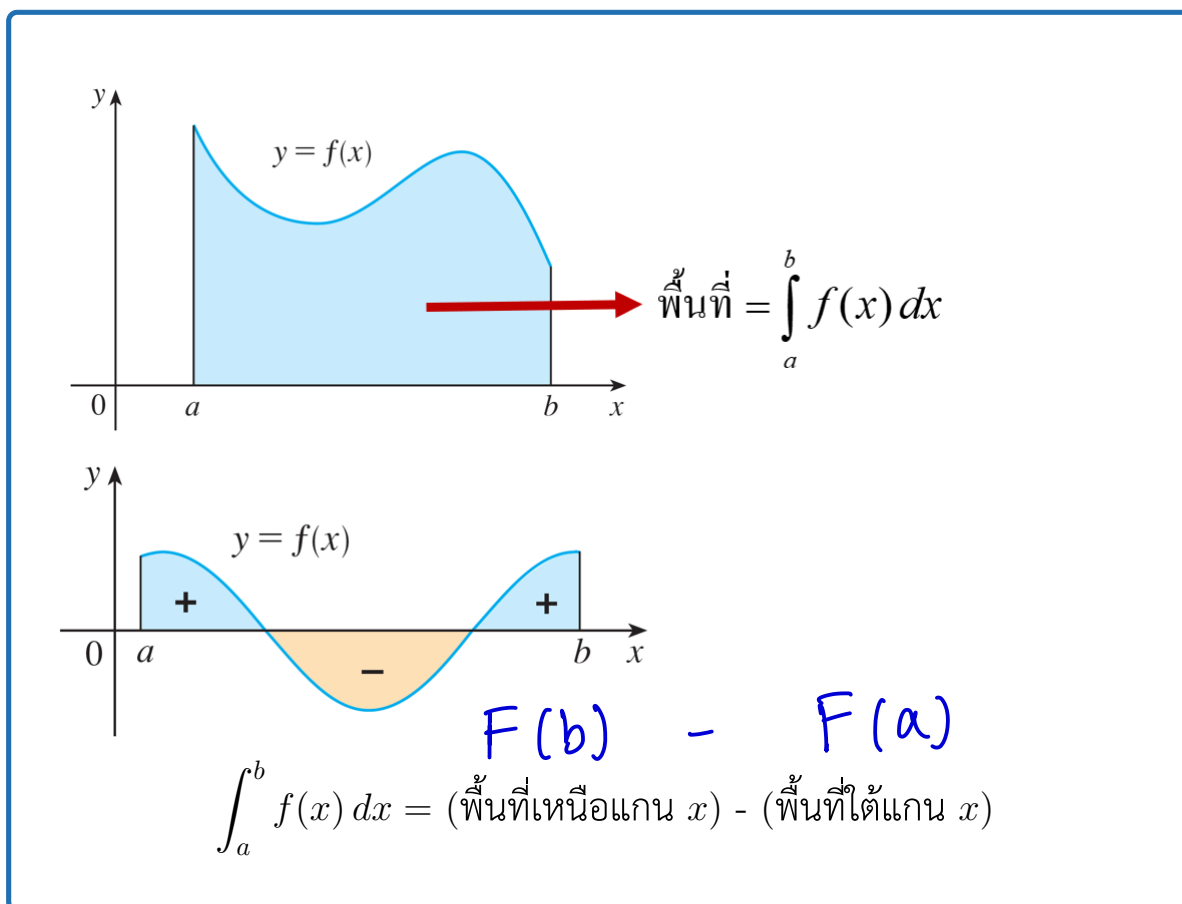
$$= \sec \theta - 2\theta + C$$

$$23. \int \sqrt{5x} dx$$

$$\frac{2\sqrt{5}}{3} x^{\frac{3}{2}} + C$$

$$24. \int \sec t (\sec t + \tan t) dt$$

$$\tan t + \sec t + C$$



$$1 \int_0^1 x^2 dx = \frac{x^3}{3} + C$$

$$= \left[\frac{(1)^3}{3} + C \right] - \left[\frac{(0)^3}{3} + C \right]$$

$$= \left(\frac{1}{3} + \cancel{C} \right) - \left(0 + \cancel{C} \right)$$

$$= \frac{1}{3}$$

$$4 \int_2^4 x dx = \frac{x^2}{2} \Big|_2^4 = \frac{(4)^2}{2} - \frac{(2)^2}{2} = \frac{16}{2} - \frac{4}{2} = \frac{12}{2} = 6$$

สมบัติของปริพันธ์จำกัดเขต

$$1. \int_b^a f(x) dx = - \int_a^b f(x) dx$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_a^a f(x) dx = 0$$

$$3. \int_a^b k dx = k(b-a) \text{ เมื่อ } k \text{ เป็นค่าคงตัว}$$

$$4. \int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$5. \int_a^b k f(x) dx = k \int_a^b f(x) dx$$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$c \in [a, b]$

$$6. \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$7. \text{ ถ้า } f(x) \geq 0 \text{ สำหรับ } a \leq x \leq b \text{ แล้ว } \int_a^b f(x) dx \geq 0$$

$$8. \text{ ถ้า } f(x) \geq g(x) \text{ สำหรับ } a \leq x \leq b \text{ แล้ว } \int_a^b f(x) dx \geq \int_a^b g(x) dx$$

25. กำหนดให้ f และ g เป็นฟังก์ชันที่หาปริพันธ์ได้ และ

$$\int_1^2 f(x) dx = -4, \quad \int_1^5 f(x) dx = 6, \quad \int_1^5 g(x) dx = 8$$

จงหาค่าของ

$$(a) \int_2^2 g(x) dx = 0 \quad \text{✗}$$

$$(b) \int_5^1 g(x) dx = - \int_1^5 g(x) dx = -8 \quad \text{✗}$$

$$(c) \int_1^2 3f(x) dx = 3 \int_1^2 f(x) dx = 3(-4) = -12 \quad \text{✗}$$

$$(d) \int_2^5 f(x) dx = \int_1^5 f(x) dx - \int_1^2 f(x) dx = 6 - (-4) = 10 \quad \text{✗}$$

$$(e) \int_1^5 [f(x) - g(x)] dx = \int_1^5 f(x) dx - \int_1^5 g(x) dx = 6 - 8 = -2 \quad \text{✗}$$

$$(f) \int_1^5 [4f(x) - g(x)] dx = 4 \int_1^5 f(x) dx - \int_1^5 g(x) dx = 4(6) - 8 = 24 - 8 = 16 \quad \text{✗}$$

26. กำหนดให้ f และ h เป็นฟังก์ชันที่หาปริพันธ์ได้ และ

$$\int_1^9 f(x) dx = -1, \quad \int_7^9 f(x) dx = 5, \quad \int_7^9 h(x) dx = 4$$

จงค่าของ

$$(a) \int_1^9 -2f(x) dx = -2(-1) = 2$$

$$(b) \int_7^9 [f(x) + h(x)] dx = 4 + 5 = 9$$

$$(c) \int_7^9 [2f(x) - 3h(x)] dx = -2$$

$$(d) \int_9^1 f(x) dx = 1$$

$$(e) \int_1^7 f(x) dx = \int_1^9 f(x) dx - \int_7^9 f(x) dx = -1 - 5 = -6$$

$$(f) \int_9^7 [h(x) - f(x)] dx$$

$$1$$

30. จงหา $\int \left(\frac{1}{x} - \frac{1}{x^2 + 1} \right) dx$

[ข้อสอบเก่า Cal 1 (1/66)]

31. จงหา $\int \tan^2 x \, dx$

$$\tan^2 x + 1 = \sec^2 x$$

[ข้อสอบเก่า Cal 1 (1/66)]

$$\tan^2 x = \sec^2 x - 1$$

$$= \int \sec^2 x - 1 \, dx$$

$$= \tan x - x + C \quad \text{✗}$$

32. กำหนดให้ $\int_0^2 f(x) \, dx = 4$, $\int_2^5 f(x) \, dx = 10$ และ $\int_2^5 g(x) \, dx = -3$

(a) จงหา $\int_2^5 [3f(x) - g(x)] \, dx$

(b) จงหา $\int_5^0 [f(x)] \, dx$

[ข้อสอบเก่า Cal 1 (1/66)]

33. จงหา $\int_{\frac{\pi}{4}}^{\frac{\pi}{4}} (\sec^2 x \cdot \tan^2 x) \, dx = 0$

[ข้อสอบเก่า Cal 1 (1/66)]

แบบฝึกหัดเพิ่มเติม

จงหาอินทิเกรตต่อไปนี้

1. $\int (x^2 - 2x + 5x^3 - 5) dx$

12. $\int \frac{(\sqrt{a} - \sqrt{x})^2}{\sqrt{x}} dx$

2. $\int \left(t^3 - \frac{2}{t^2} \right) dt$

13. $\int \frac{x dx}{(a + bx^2)^3}$

3. $\int \frac{(\sqrt[3]{u} + 1)^3}{\sqrt{u}} du$

14. $\int y(a - by^2)^2 dy$

4. $\int (\theta + 3 \cot^2 \theta) d\theta$

15. $\int (u + 4)(2u + 1) du$

5. $\int (\sec \varphi + 2 \tan \varphi)^2 d\varphi$

16. $\int \sqrt{t}(t^2 + 3t + 2) dt$

6. $\int (1 + \sin^2 \varphi \csc \varphi) d\varphi$

17. $\int \frac{1 + \sqrt{x} + x}{x} dx$

7. $\int \frac{\sin^2 z}{1 + \cos z} dz$

18. $\int \left(x^2 + 1 + \frac{1}{x^2 + 1} \right) dx$

8. $\int \frac{4x^2 - 2\sqrt{x}}{\sqrt{x}} dx$

19. $\int \left(\frac{1+r}{r} \right)^2 dr$

9. $\int \left(\frac{x^2}{2} - \frac{2}{x^2} \right) dx$

20. $\int (2 + \tan^2 \theta) d\theta$

10. $\int x(2x + 1)^2 dx$

21. $\int 2^t(1 + 5^t) dt$

11. $\int \sqrt{x}(3x - 2) dx$

22. $\int \frac{\sin 2x}{\sin x} dx$